MG MUSALLAM GROUP · DELIVERY OPERATIONS

MGHub

System Architecture & Technical Overview

The platform end to end — business modules, the Firestore data model, the payroll and synchronisation engines, every callable, route, cron and scheduled job, and the feature matrix that maps them to the screens operators use.

Next.js 16 · Firestore | Cloud Functions gen2 · me-central1 | Cloudflare DO · R2

Read from the MG-GO-Admin and MG-GO source trees · No placeholders · English edition

Version 1.0 · 13 Nov 2026

MGHUB · SYSTEM ARCHITECTURE & TECHNICAL OVERVIEW

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VERSION
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READER
Client technical stakeholders, architects, onboarding engineers

REBUILD
npm run docs:architecture

Platform: MGHub Admin & Operations Portal (Musallam Group / DPD) **Document version:** 1.0 · **Generated:** 13 November 2026 **Sources:** `src/app`, `src/features`, `src/lib`, `functions/src`, `infra/workers/dpd-live`, `supabase/migrations`, `vercel.json` **Method:** every table, route, function and cron below was read from the repository. Nothing is illustrative.

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1 Executive Summary & Core Modules

1.1 What the platform is

MGHub is the back-office control tower for a contracted rider workforce: roughly 950 delivery riders across Kuwait, dispatched by third-party partners (Americana, Keeta, Talabat, Uber Eats, Door Dash, Five Guys). **Order allocation is external** — the partner apps decide which rider gets which order. MGHub owns everything around that:

- who the rider is (identity, documents, contracts, residency)
- whether they were present and productive (shift, attendance, GPS, zone discipline)
- what they are paid (per-order incentives, monthly payroll, HR payroll)
- what the client can see (a scoped partner portal)
- the operational surfaces built on top (e-signature, visit bookings, accident cases, fleet maintenance, notifications)

Two clients consume it:

Client	Repository	Surface
Admin / staff	`MGgo-Admin` (this repo)	Full back-office web app
Rider	`MG-GO` (separate repo)	Flutter app, OTP + passcode login

A third surface — the Cloudflare Worker `dpd-live` — sits between the rider phones and the admin live map.

1.2 Business modules (route inventory, from `src/app/[locale]/(dashboard)`)

#	Module	Route	Permission	Purpose
1	**Dashboard / Launcher**	`/dashboard`	`dashboard.view`	Dark launcher grid; the hub of the app
1b	Control Tower	`/dashboard/ops`	`dashboard.view`	KPI snapshot across modules
2	**Live Tracking V1**	`/live-tracking`	`live_tracking.view`	Durable pin per rider from `driver_locations`
2b	**Live Tracking V2**	`/live-tracking-v2`	`live_tracking.view`	1 Hz deck.gl canvas over a Cloudflare Durable Object
3	**Employees**	`/drivers`, `/drivers/[id]`, `/drivers/new`, `/drivers/groups`, `/drivers/dashboard`	`drivers.view` / `drivers.manage`	Rider roster, intake → approval, documents, device link

#	Module	Route	Permission	Purpose
4	**Fleet Hub**	`/fleet` → `/vehicles`, `/fuel`, `/assets`, `/fleet/service`, `/fleet/handover`, `/fleet/accident`, `/fleet/rj`	`vehicles.*`, `fuel.*`, `assets.*`, `service.*`	Vehicles, fuel ledger, assets, workshop, handovers, accidents
5	**Deliveries**	`/deliveries`, `/deliveries/reconciliation`, `/deliveries/zones`	`deliveries.view` / `deliveries.manage`, `order_recon.*`	Verify / reject orders, AM-vs-MGGO comparison
6	**Payroll Hub**	`/payroll-hub` → `/payroll`, `/payroll/combined`, `/payroll/attendance-orders`, `/payroll/hr`, `/payroll/settings`, `/payroll/pay-plan`	`payroll.*` (10 slugs)	Monthly payroll, HR sheet, rules engine
7	**Operations Hub**	`/operations` → `/operations/roster` (+ `/roster/upload`, `/roster/changes`, `/roster/alerts`, `/roster/report`)	`attendance.*`, `earnings.*`, `drivers.manage`	Rule tables, shift roster, attendance alerting
8	**Attendance**	`/attendance`	`attendance.view` / `attendance.manage`	Live presence, logs, exceptions, analytics
9	**Notifications**	`/notifications` (+ `/new`, `/[id]`, `/history`, `/automations`, `/templates`, `/analytics`)	`notifications.*` (5 slugs)	Campaigns, audiences, FCM dispatch, automations
10	**Roles & Permissions**	`/settings/roles`, `/settings/staff-access`	`roles.manage`, super admin	4-level access editor + role matrix
11	**EmployeeDesk**	`/employeedesk` (+ `/esign/*`, `/visits/*`, `/all`, `/[id]`, `/reports`, `/settings`)	`requests.*`, `esign.*`, `visits.*`	Request & Complaint (RCM), e-signature, visit bookings
12	**Visit Slots**	`/visit-bookings` (+ `/all`, `/calendar`, `/reception`, `/slots`, `/departments`, `/branches`, `/reports`)	`visits.*` (4 slugs)	Appointment booking for HR/compliance desks
13	**Performance**	`/performance`, `/performance/settings`	`performance.*` (5 slugs)	SOP v2.0 OPD/DPD/UTR analytics

#	Module	Route	Permission	Purpose
14	**Client Portal**	`/client-portal-v2` (+ `/employees`, `/clients`, `/new`, `/preview/[projectKey]`)	`client_portal.view`	Partner-scoped read-only dashboard
15	**System Settings**	`/settings` (+ 24 sub-pages)	`settings.*`	Branding, menus, languages, catalogs, storage, cleanup
16	Support / Assistant / Compliance / Wrong actions / Document expiry / Driver devices / Payouts / Earnings / DPD rules	`/support`, `/assistant`, `/compliance`, `/wrong-actions`, `/document-expiry`, `/driver-devices`, `/payouts`, `/earnings`, `/delivery-rules`, `/incentive-rules`	per module	See §5

41 top-level route folders under `src/app/[locale]/(dashboard)`, **92 permission slugs** in `src/lib/auth/permission-catalog.ts`.

1.3 Tech stack

Layer	Choice	Where
Framework	Next.js **16.2.6**, App Router, Turbopack, React Server Components	`package.json`, `next.config.ts`
Language	TypeScript (strict), no `.js` app sources	`tsconfig.json`
Styling	Tailwind CSS v4 + shadcn/ui primitives on CSS custom properties; single locked navy/green theme	`src/app/globals.css`, `src/app/themes.css`, `src/components/ui/*`
Product components	`src/components/app/*` (`AppPage`, `AppListCard`, `AppDataTable`, `AppModalFooter`, `TableToolbar`, `AppHubShell`)	shared shell for every page
Server state	**TanStack Query v5** — one `QueryClient` per browser tab; `staleTime` 60 s, `gcTime` 5 min, `refetchOnWindowFocus: false`	`src/lib/query/*`, `src/components/providers/query-provider.tsx`
Client state	React hooks + Riverpod-equivalent pattern via context providers; no Redux	`src/contexts/*`
i18n	**next-intl**, locales `en` + `ar` (RTL), **12,625 keys × 2** with a parity guard	`src/messages/*.json`, `scripts/check-messages.mjs`
Database	**Firestore** (named database `default`), Admin SDK server-side only	`src/lib/firebase/*`, `functions/src/core/fs.ts`
Auth	Firebase Auth → **session cookie** (`__session`) verified in `src/proxy.ts`; custom claims carry `superAdmin`	`src/lib/firebase/session.ts`, `src/proxy.ts`
Backend	**Cloud Functions gen2** (`firebase-functions/v2`) in `me-central1` / `musallam-delivery-prod`	`functions/src`

Layer	Choice	Where
Real-time (maps)	**Cloudflare Durable Object** `FleetRoom` + WebSocket, with a Supabase-Broadcast-era poll fallback	`infra/workers/dpd-live`
Object storage	**Cloudflare R2** `dpd-private-prod` (driver docs, e-sign PDFs, fuel, request attachments, GPS archive)	`src/lib/storage/r2-client.ts`
Hosting	**Vercel**, region `bom1`, `maxDuration` 300 s on the dashboard layout, the order-recon routes and `/deliveries/reconciliation`	`vercel.json`, `(dashboard)/layout.tsx`
Charts	**Recharts** (Performance trend only) + hand-built CSS/SVG bars elsewhere	`src/features/performance`
Maps	Google Maps JS + `@vis.gl/react-google-maps`; **deck.gl** for V2; Leaflet for `/zones` editor	`src/features/driver-tracking`, `src/features/live-tracking-v2`, `src/features/zones`
Excel / CSV	**ExcelJS** (styled workbooks) + **xlsx** (namespace import) + custom CSV writers	`src/lib/*xlsx*`, `src/lib/csv/*`
Push	**Firebase Cloud Messaging** (FCM) via Admin SDK	`functions/src/rpcs/driver-push-tokens.ts`, `notifications.ts`
PDF	Chrome print for e-sign sheets; `pdf-lib` for signature stamping	`functions/src/rpcs/esign-compose*`
Observability	Sentry (client + functions), `perf` JSON logs from the callable router	`src/lib/sentry/*`, `functions/src/perf/*`

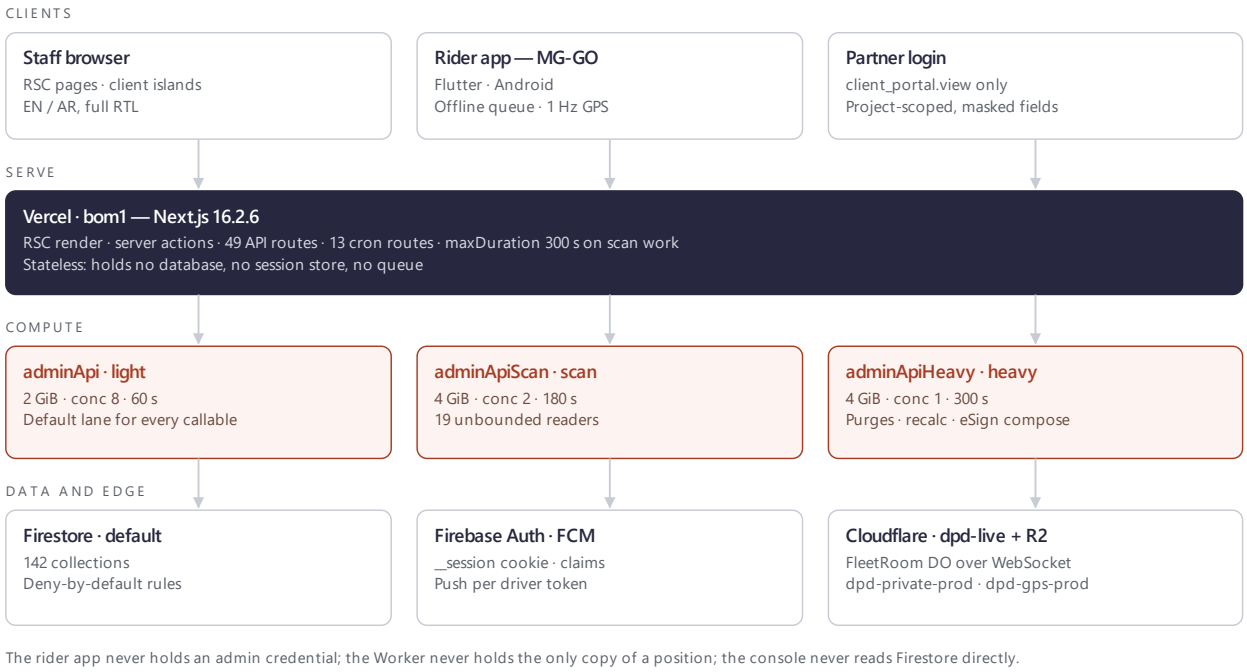


Figure 1. The platform in four layers. Every arrow is a real network hop: an HTTPS callable, a WebSocket frame, or an Admin SDK write.

2 Database Schema & Data Models

2.1 Two schemas, one live

The project migrated from **Supabase Postgres** to **Firestore**. Both exist in the tree; only one is live.

Schema	Status	Location	Count
Firestore collections	**Live** — every read and write goes here	`src/lib/firebase/db.ts`, `functions/src/core/collections.ts`	**142 named collections** (+ 6 `uniq_*` lock collections, `counters`)
Postgres migrations	**Historical** — the Supabase project `eoksxkdssptgyqyywdju` is paused; admin and Worker no longer call it	`supabase/migrations/*.sql`	398 files

> **Read this before writing a query.** A Postgres table name in a migration does **not** mean a live table. `deliveries`, `drivers`, `requests`, `esign_requests` and the rest are Firestore collections now. `src/types/database.ts` is a generated artefact kept for type continuity, not a live schema.

2.2 Identity, access and configuration

Collection	Purpose	Key fields	Relationships
`profiles`	Auth mirror + panel access	`id` (= `auth.uid`), `role` (`staff` | `rider`), `approval_status` (`pending` | `approved` | `rejected`), `admin_role_id`, `access_kind` (`manager` | `user` | `partner`), `partner_project_key`, `staff_department`, `sessions_revoked_at`, `archived_at`	1:1 `drivers`; N:1 `admin_roles`
`drivers`	The rider record	`employee_id` (unique, 4–8 digits), `driver_code` (5 digits), `client_id`, `phone`, `civil_id`, `passport_no`, `zone_id`, `partner_id`, `vehicle_id`, `project_key`, `status`, `app_passcode`, `custom_fields` (jsonb), `current_odometer_km`, `avatar_object_key`	N:1 `zones`, `partners`, `vehicles`; M:N `restaurants` via `driver_restaurants`
`driver_intakes`	Add-Driver draft before approval	same identity fields + `linked_profile_id`, `archived_at`	1:1 `drivers` after approve
`admin_roles`	Panel role	`slug`, `name`, `is_super_admin`	1:N `admin_role_permissions`
`admin_permissions`	Slug catalog (172 slugs) + `catalog` doc	`slugs: string[]`	—

Collection	Purpose	Key fields	Relationships
`admin_role_permissions`	Role → slugs	`permission_slugs: string[]`	N:1 `admin_roles`
`admin_user_permissions`	Per-user ticks	`permission_slugs: string[]`	1:1 `profiles`
`admin_ui_preferences`	Per-user column order / view	`user_id`, `preference_key`, `value` (`order`, `visible`, `sort`, `groupBy`, `viewMode`)	N:1 `profiles`
`admin_role_ui_defaults`	Per-role column defaults	`role_id`, `preference_key`, `value`	N:1 `admin_roles`
`admin_activity_logs`	Audit trail	`action`, `entity_type`, `entity_id`, `before`, `after`, `actor_id`, `page_path`	N:1 `profiles`
`auth_epochs`	Cache-bust counters (`current`)	`access`, `settings`, `menu`, `payroll`, …	—
`menu_configs`	Per-role sidebar tree	`admin_roles.slug` → JSON tree	N:1 `admin_roles`
`locales`	Enabled locales	`code`, `is_enabled`, `is_default`, `dir`	—
`app_settings`	Singleton doc `1`	branding (`app_name`, `logo_url`, `font`), `maintenance_mode`, `super_admin_claimed`, `super_admin_user_id`, force-update block, `pickup_auto_cancel_hours`, `attendance_auto_checkout_minutes`, `delivery_ontime_minutes`, `performance_score_weights`, `incentive_band_math_from`	—
`counters`	8 code sequences	`driver_code_seq`, `restaurant_code_seq`, `request_code_seq`, `visit_booking_code_seq`, `esign_code_seq`, `appointment_code_seq`, `fuel_refund_code_seq`, `esign_batch_code_seq`	—
`uniq_employee_id` · `uniq_phone` · `uniq_civil_id` · `uniq_passport` · `uniq_plate` · `uniq_passcode`	Lock documents standing in for partial unique indexes	one doc per held value	—

Constraints that live in code, not in a schema (Firestore has no DDL):

Rule	Enforced at
`employee_id` unique among live riders, 4–8 digits	`driver-uniqueness.ts`, `uniq_employee_id`, `admin_approve_driver`

Rule	Enforced at
`phone` / `civil_id` / `passport` optional but unique when present	partial-index lock docs; `''` unrepresentable
`driver_code` exactly 5 digits from `driver_code_seq`	`driver-code.ts`
`app_passcode` 6 digits, minted when `status` becomes `active`	`admin_approve_driver`
Kuwait day stored as `YYYY-MM-DD` string, never derived at read	`kuwaitDayString()` in `db.ts`

2.3 Workforce, operations and delivery

Collection	Purpose	Key fields	Relation ships
`partners`	Dispatch partners = **Clients** in the UI	`name`, `slug`, `logo_url`, `description`	1:N `restaurants`, `drivers`
`restaurants`	Merchant store	`name`, `partner_id`, `zone_id`, `latitude`, `longitude`, `status`, `external_merchant_id`	N:1 `partners`, `zones`; M:N `drivers`
`driver_restaurants`	Live rider ↔ store assignment (M:N)	`driver_id`, `restaurant_id`	junction
`driver_intake_restaurants`	Intake-stage assignment	same	junction
`restaurant_geofences`	Per-store inclusion / exclusion shapes	`restaurant_id`, `kind`, `zone_type`, `geometry`	N:1 `restaurants`
`zones`	Delivery areas	`name`, `code`, `zone_type`, `geometry` (GeoJSON polygon/circle), `color`	1:N `restaurants`, `drivers`
`vehicles`	Fleet record	`reg_number` (plate = Vehicle ID, unique), `bike_id` (slug), `vehicle_type_key`, `vehicle_company_id`, `vehicle_use_type`, `condition`, `status`, `current_driver_id`, `current_odometer_km`, `chassis_no`, `model_year`, `company_id`	1:1 `drivers` (current holder)

Collection	Purpose	Key fields	Relationships
`vehicle_types` · `vehicle_companies` · `vehicle_use_types`	Fleet catalogs	`key`, `label_en`, `label_ar`, `is_active`	—
`deliveries`	One order	`external_order_id`, `driver_id`, `restaurant_id`, `zone_id`, `partner_id`, `status` (`pending` | `in_transit` | `pending_verification` | `verified` | `rejected` | `cancelled`), `created_at`, `pickup_at`, `started_to_restaurant_at`, `delivered_at`, `shift_date`, `proof_keys`, `rejection_reason`	N:1 `drivers`, `restaurants`
`rollups_driver_day`	Pre-aggregated per rider-day	`{driverId}_{YYYY-MM-DD}`, `pending`, `in_transit`, `cancelled`, `attendance_present`, `zone_id`, `partial`	derived
`rollups_zone_month`	Pre-aggregated per zone-month	`{zoneId}_{YYYY-MM}`, orders, rider days	derived
`attendance_logs`	One check-in	`driver_id`, `log_date`, `check_in_at`, `check_out_at`, `check_out_reason` (`manual` | `auto_shift_end` | `auto_timeout`), `status`, denormalised identity	N:1 `drivers`
`driver_sessions`	Duty session	`driver_id`, `is_online`, `went_online_at`, `ended_at`, `device_id`	N:1 `drivers`
`driver_daily_shifts`	Rider-submitted or roster-uploaded shift	`driver_id`, `shift_date`, `start_at`, `end_at`, `source` (`roster` | `driver`), `approved_change_id`	N:1 `drivers`
`roster_changes`	Requested schedule change	`driver_id`, `shift_date`, `status`, `decided_by`	N:1 `driver_daily_shifts`
`driver_off_structure`	Monthly OFF entitlement	`{driverId}_{YYYY-MM}`, `off_days` (default 2)	N:1 `drivers`
`driver_leave_plans`	Leave ledger	`driver_id`, `leave_type`, `start`, `end`, `days`, `status`, `year_month`, `source`	N:1 `drivers`

Collection	Purpose	Key fields	Relation ships
`driver_locations`	Latest fix, one doc per rider	`driver_id` (PK), `lat`, `lng`, `speed_mps`, `heading_deg`, `accuracy_meters`, `zone_status`, `out_of_zone_since`, `last_seen_at`, `last_report_at`, `distance_today_meters`, `day`	1:1 `drivers`
`driver_location_events`	90-day position history	append-only	N:1 `drivers`
`driver_operation_events`	Server-authored audit of every rider RPC	`driver_id`, `event_type`, `payload`, `expire_at` (30 d)	N:1 `drivers`
`driver_telemetry_events`	Client-authored phone telemetry	`event_id` (idempotency), `event_type`, sanitised `context`, `expire_at` (14 d)	N:1 `drivers`
`rider_day_stats`	Per-day GPS totals	`{driverId}_{day}`	—
`fleet_events`	Edge-derived Class B events (`overspeed`, `sustained_idle`, `low_battery`, `zone.exit`, `range.exit`, `gps.offline`, `shift.late`, `shift.overrun`)	`driver_id`, `type`, `at`, `expire_at` (400 d)	N:1 `drivers`
`rider_signals`	Screen-refresh counters for the app	`{uid}` → `home_rev`, `earnings_rev`, …	1:1 `drivers`
`driver_push_tokens`	FCM tokens	`driver_id`, `token`, `is_active`, `last_seen_at`	N:1 `drivers`
`driver_device_sessions`	Device binding + build reporting	`driver_id`, `device_id`, `app_version`, `device_meta`, `device_meta_at`	N:1 `drivers`
`driver_shifts` (via `driver_daily_shifts`) · `driver_devices` (via `driver_device_sessions`)	see above		

2.4 Money: earnings, payroll, recon

Collection	Purpose	Key fields	Relationships
`delivery_rules`	Per-delivery incentive rule	`name`, `partner_id` / `restaurant_id` / `zone_id`, `dpd_target`, `dpd_period`, `start_date`, `end_date`, `reward_mode`, `payout_mode`, `pdp_targets`, `priority`, `is_active`	1:N `delivery_rule_scopes`, `incentive_rule_tiers`
`delivery_rule_scopes`	Multi-scope rule targeting	`rule_id`, `scope_type`, `scope_id`	N:1 `delivery_rules`
`incentive_rules`	Tiered / cumulative plans	`overrides_others`, `priority`, `tiers`, `applies_to`	1:N `incentive_rule_tiers`, `incentive_rule_scopes`
`incentive_rule_tiers` · `incentive_rule_scopes`	tier rows / scope rows	—	N:1
`driver_earnings_daily`	Priced per rider-day	`{driverId}_{earn_date}`, `deliveries`, `incentive_kwd`, `breakdown`, `calculated_at`	N:1 `drivers`
`driver_wallet_entries`	Ledger	`driver_id`, `kind`, `amount_kwd`, `ref`	N:1 `drivers`
`driver_payouts` · `payout_runs`	Payout batches	`run_id`, `driver_id`, `amount`	N:1
`payroll_clients`	One row per payroll client (Americana, Keeta, …)	`key`, `display_name`, hour values, thresholds, `zone_efficiency_good` / `_low`, `employment_category`, `rules`	1:N `payroll_client_rules`
`payroll_client_rules`	Ordered rule engine rows	`conditions` (jsonb), `all`, `result`	N:1 `payroll_clients`
`payroll_zone_metrics`	Previous-month zone efficiency	`{zoneId}_{YYYY-MM}`, `orders`, `rider_days`, `dpd`, `target_dpd`, `efficiency_pct`, `dpd_used`, `target_used`, `category`	N:1 `zones`
`payroll_zone_settings`	Global + per-zone thresholds	`target_dpd`, `good`, `low`, per-zone overrides	—
`payroll_manual_adjustments`	Append-only hand edits	`driver_id`, `date`, `status`, `hours`, `reason`, `actor_id`, `undo_batch`	N:1 `drivers`
`payroll_day_hours`	Loaded hours for a Keeta/Five Guys day	`{driverId}_{YYYY-MM-DD}`	N:1 `drivers`
`payroll_order_overrides`	Manual order count per day	`{driverId}_{YYYY-MM-DD}`, `orders`	—

Collection	Purpose	Key fields	Relationships
`payroll_rider_statuses` (`roster` doc)	Status label catalog	`rosters`, `customs`	—
`payroll_column_config`	Org-wide column labels + per-view visibility	`views`, `labels`	—
`payroll_rule_audit_logs`	Rule change audit	`before`, `after`, `actor_id`	—
`rider_compliance_daily`	Invalid / rejected / cancelled days	`{driverId}_{day}`	N:1 `drivers`
`order_recon_runs`	One applied AM sheet	`status`, `from_date`, `to_date`, `kpi`, `created_at`	1:N `order_recon_rows`
`order_recon_rows`	Melted AM cells	`run_id`, `employee_id`, `work_date`, `excel_orders`, `app_orders`, `difference`	N:1 `order_recon_runs`
`fuel_fills`	Fuel ledger	`driver_id`, `vehicle_id`, `litres`, `cost_kwd`, `fuel_type`, `transfer_type`	N:1
`fuel_withdrawn_overrides`	Per-month withdrawn flag	`{driverId}_{vehicleId}_{YYYY-MM}`	—

2.5 EmployeeDesk: requests, e-signature, visits

Collection	Purpose	Key fields	Relationships
`requests`	RCM case	`request_code` (`RCM-####`), `request_type` (FK → `request_type_definitions.key`), `status`, `driver_id`, `payload` (jsonb), `current_step_order`, `attention_at`, `acknowledged_at`, `sla_due_at`, `closed_at`, `fuel_transfer_type`, `vehicle_id`, `is_confidential`	1:N `request_approval_steps`, `request_attachments`, `request_clarifications`
`request_type_definitions`	Type catalog (8 system + custom)	`key`, `is_system`, `terminal_status`, `requires_ack`	1:N `request_field_definitions`
`request_field_definitions`	Per-type field schema	`type_key`, `key`, `field_type`, `options`, `is_required`, `is_server_required`, `min_value` / `max_value`, `visibility`	N:1

Collection	Purpose	Key fields	Relationships
`request_approval_step_templates`	Chain definition	`type_key`, `step_order`, `role_key`, `sla_minutes`, `breach_action`	1:N `request_approval_steps`
`request_approval_steps`	Live step	`request_id`, `step_order`, `status`, `decided_by`, `actor_display_name`, `sla_due_at`	N:1 `requests`
`request_attachments` · `request_clarifications` · `request_comments` · `request_forwards`	Case children	—	N:1
`request_staff_access` · `esign_staff_access`	Per-type sender/receiver grants	`type_key`, `can_view`, `can_decide`	—
`esign_templates` · `esign_template_fields`	Template + field map	`kind` (`penalty` | `loan` | `payslip` | `general_doc` |…), `sheet_html`, `sheet_render_version`, `field_values`	1:N
`esign_categories`	Category catalog incl. `parent_key`	`key`, `screenshot_restricted`, `is_active`	self 1:N
`esign_requests`	One signature request	`request_code` (`SIG-####`), `category_key`, `status`, `sent_at`, `viewed_at`, `declined_at`, `declaration_accepted_at`, `document_storage_key`, `signed_document_storage_key`, `employee_snapshot`, `field_values`	1:N `esign_request_signers`, `esign_request_attachments`
`esign_request_signers`	Multi-signer	`request_id`, `driver_id` **xor** `staff_user_id`, `sign_order`, `signed_at`	N:1
`esign_batches` · `esign_batch_rows`	Bulk send (`BAT-####`)	`template_id`, `source_filename`, `total_count`, `progress`	1:N
`esign_drafts` · `esign_audit_events`	Drafts / audit	—	—
`visit_branches`	Location	`key`, `name`, `is_default`, `working_dows`, `opening_time`, `closing_time`, `lunch_start` / `lunch_end`	1:N `visit_departments`, `visit_slots`
`visit_departments`	Desk	`key`, `label_en`, `label_ar`, `branch_id` (null = all), `desks_count`, `avg_handling_minutes`, `is_active`	N:1 `visit_branches`

Collection	Purpose	Key fields	Relationships
`visit_slots`	Availability	`department_key`, `branch_id`, `weekday`, `start_time`, `end_time`, `capacity`	N:1
`visit_bookings`	Booking (`VIS-#####`)	`booking_code`, `driver_id`, `department_key`, `branch_id`, `slot_id`, `scheduled_date`, `status`, `checked_in_at`, `note`, `note_to_rider`, `vehicle_id`	N:1
`visit_booking_notes` · `visit_blocked_dates`	Admin thread / blackout dates	—	—

2.6 Notifications, performance, fleet service

Collection	Purpose	Key fields
`notification_campaigns`	Campaign	`title`, `body`, `status` (`draft` | `scheduled` | `sent` | `failed` | `partially_failed`), `target_spec`, `audience_snapshot_id`, `screenshot_restricted`, `track_engagement`, `created_at`
`notification_templates`	Reusable body + action preset	`action_type`, `fields`, `record_type`, `is_archived`
`notification_audience_snapshots`	Frozen audience	`ids[]`, `count`, `spec`
`notification_dispatch_runs` · `notification_dispatch_items`	Per-campaign send ledger	`campaign_id`, `driver_id`, `status`, `dismissed_at`, `opened_at`
`notification_automations` · `notification_automation_events`	Trigger → send rules	`kind`, `lead_days`, `is_active`
`notification_client_events` · `notification_events` · `notification_analytics_daily` · `notification_timeline`	Engagement	`{day}_{campaignId}`, delivery / open / click / failure
`notification_device_tokens` · `driver_push_tokens`	FCM registry	`token`, `is_active`, `last_seen_at`
`driver_performance_daily`	**Raw numerators and denominators**, never a score	per rider-day component inputs
`performance_score_components`	Weighted components (`punctuality`, `duty_ratio`, `on_time`, `speed`, `zone`, `gps`, `conduct`)	`key`, `weight`, `sort_order`, `is_active`

Collection	Purpose	Key fields
`performance_rating_criteria`	Team × criterion	`team_key` , `key` , `weight`
`driver_performance_ratings`	`(driver_id, criterion_id, period_month)`	`score` 1–5
`driver_performance_rating_notes`	`(driver_id, team_key, period_month)`	`note`
`performance_rating_team_members`	Who may rate which team	`team_key` , `user_id`
`vehicle_services`	Workshop ledger (legacy + additive)	`vehicle_id` , `service_type_key` , `odometer_km` , `cost_kwd` , `stage` , `parts[]`
`service_types`	Service Type catalog: **468 part rows** + 5 maintenance programs, 12 categories	`key` , `label_en` , `label_ar` , `category` , `tracks_maintenance`
`vehicle_odometer_readings`	One reading per service or manual edit	`vehicle_id` , `km` , `source` (`dashcam` | `manual` | `service`), `previous_km` , `lowered` , `reason`
`vehicle_maintenance_schedules`	Rule per vehicle / make / make-model / type	`scope` , `km_interval` , `day_interval` , `alert_km` , `alert_days`
`vehicle_maintenance_exclusions` · `_overrides`	Opt-out / hand-set next service	`{vehicleId}_{programKey}`
`vehicle_day_distance`	Bike daily distance	`{vehicleId}_{day}` , `total_meters` , `by_driver` , `sessions`
`service_maintenance_settings` · `service_odometer_sync`	Singleton `current` docs	provisional intervals / dashcam config
`vehicle_accidents`	Accident case	`reference` (`ACC-####`), frozen vehicle + driver snapshot, `stage` , `cost_kwd` , `attachment_keys[]`
`asset_catalog` · `asset_assignments`	Inventory + issued assets	`category` , `penalty_kwd` , `total_quantity` , `quantity`
`custom_field_definitions`	**Admin-defined columns**	`entity_type` (`driver` | `partner` | `restaurant` | `zone` | `vehicle`), `key` , `label` , `field_type` , `options` , `required` , `is_active` , `sort_order`
`document_tracking`	Expiry per document	`driver_id` , `doc_type` , `expiry_date` , `lead_days`
`wrong_actions`	Conduct incidents	`driver_id` , `action_type` , `severity` (`low` | `medium` | `high`), `occurred_at`
`order_exceptions`	Rider-filed rejected / cancelled order	`driver_id` , `kind` , `source` , `order_id` , `reason` , `proof_keys[]` , `status` , `day`

Collection	Purpose	Key fields
`driver_groups` · `driver_group_members`	Grouping	`member_ids[]` mirrored
`source_companies` · `employee_statuses` · `employee_sections` · `nationality_categories` · `bank_codes` · `job_positions` · `sponsorships`	Employee master catalogs	`key` , `label_en` , `label_ar` , `is_active` , `sort_order`
`support_threads` · `support_messages` · `appointment_slots` · `appointments`	Support + meetings	—
`client_portals`	One portal config per project key	`project_key` , `theme` , `rate_cards`
`driver_import_batches` · `driver_import_rows`	Bulk employee import job	`status` , `heartbeat_at` , `pending_store` , `uploaded_by`
`page_read_snapshots` · `page_read_generations`	Gzipped read-through cache for scan-lane pages	`packed` , `generation` , `fresh_until`
`admin_cost_samples`	Sampled cost telemetry	—
`admin_allowlist`	Sign-up allowlist	`email`

2.7 Entity-relationship diagram (core)

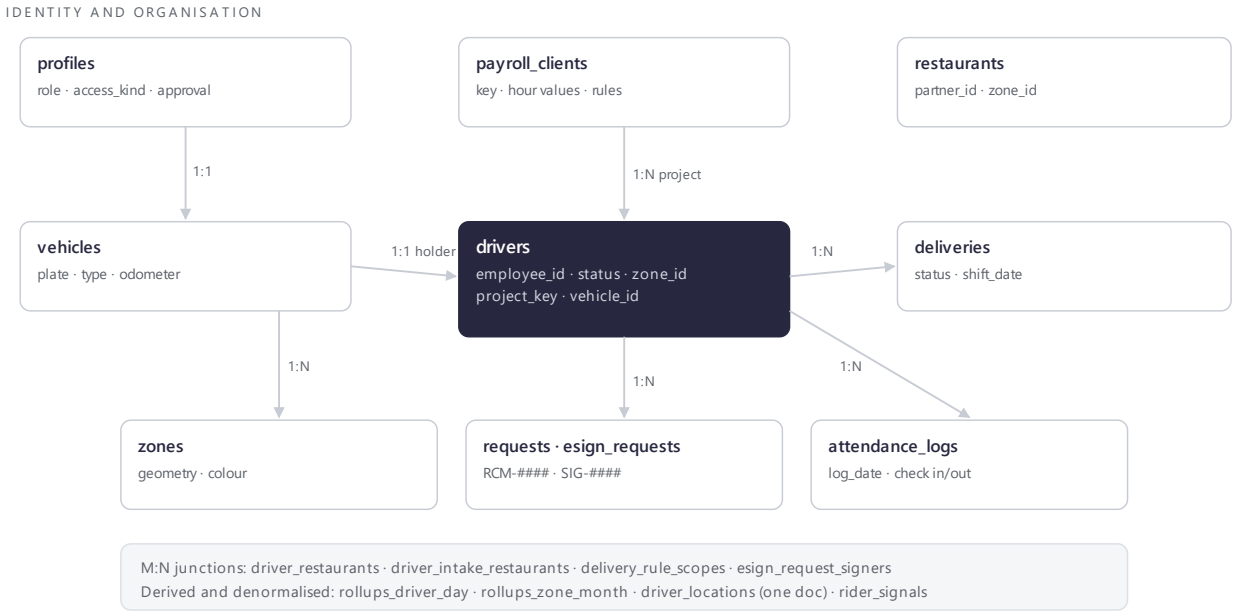


Figure 2. The core graph. Everything the console lists hangs off drivers; everything the rider files hangs off drivers too, which is why one identity fix corrects a dozen screens.

3 Core Application Functions & Business Logic

3.1 Session, RBAC and permission resolution

The chain is deliberately short — a page render costs a fixed number of reads, not one per permission.

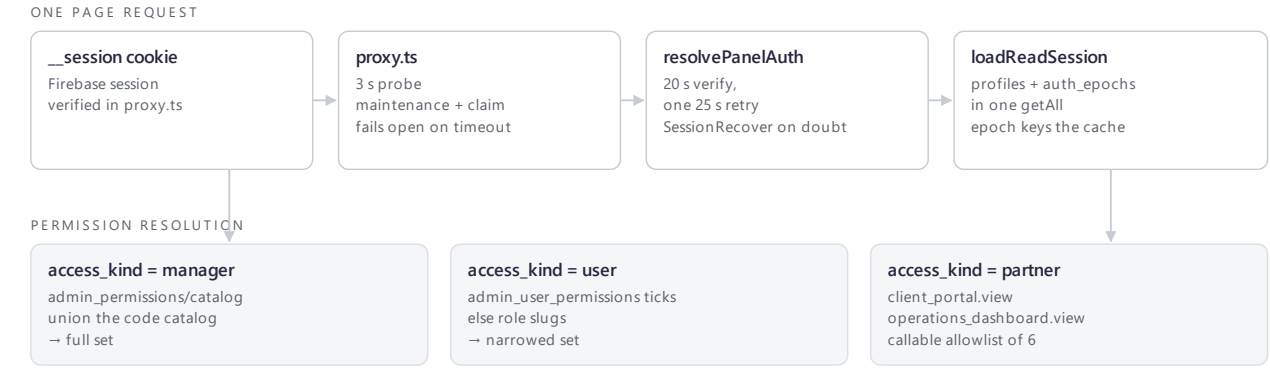


Figure 3. Session to permission in one pass. A slow or unreachable Auth backend resolves to SessionRecover — an unproven session is never treated as a signed-out one.

Function	File	Behaviour
`panelGate`	`src/lib/auth/staff-entry.ts`	Pure state machine: `login` / `not_authorized` / `claim` / `pending` / `maintenance` / `unavailable` / `ok`
`resolvePanelAuth`	`src/lib/auth/require-permission.ts`	Returns `SessionRecover` instead of throwing when auth is unproven — a slow GoTrue is not a logout
`requirePermission` / `requireAnyPermission` / `requireSuperAdmin` / `requireNotPartner`	same	Page guards; redirect to `/unauthorized`
`loadReadSession`	`src/lib/auth/epoch-session.ts`	`auth_epochs/current` + `profiles/{uid}` in **one** `getAll` ; permission docs re-read against the epoch on a miss
`enrichSessionPermissions`	`src/lib/auth/profile-auth.ts`	Manager = catalog ∪ code catalog (`catalogWithCodeSlugs`); User = ticks; Partner = 2 fixed slugs
`permissionGrantedByTicks`	`src/lib/auth/staff-access.ts`	`*.manage` ↔ create/edit/delete alias; `bulk_delete` **never granted by another tick**
`hasPermissionInSet`	`src/lib/auth/permissions.ts`	Super admin always passes

The four access levels on `/settings/roles` (`src/lib/auth/app-access.ts`) map onto existing slugs — no parallel permission model:

Level	Effect
No access	No slug
Viewer	`*.view`
User	create/edit/delete
Manager	`*.manage` + `*.bulk_delete`

Epoch caches. `auth_epochs/current` holds counters (`access` , `settings` , `menu` , `payroll` , …). Every role, tick, catalog, branding and menu write goes through `writeWithEpoch` , which increments the counter in the same transaction; readers key their cache on it. A change therefore lands on the next request, with a 5-minute TTL as the safety net. `clearCatalogSlugsCache()` is the test seam.

Partner scoping. `access_kind: 'partner'` is locked to `partner_project_key` ∈ { `keeta` , `americana` , `unassigned` , `five_guys` } and only two slugs. `requireStaff` refuses every callable outside `PARTNER_CALLABLE_ALLOWLIST` (6 names), and masked fields (phone, bank, salary, leave reason) are dropped server-side in `functions/src/rpcs/*` — never in the browser.

3.2 Catalog synchronisation (Employees ↔ HR Payroll ↔ Profiles)

The client's 5.2 ask — "a change made once reflects everywhere" — is implemented as **one catalog, many readers**, not as a sync job.

Catalog	Collection	Written by	Read by
Partners / Clients	`partners`	`/partners` , OperationsHub	Employees form (Client dropdown), Payroll Client Name, Client Portal, Incentive rule scope
Restaurants	`restaurants`	`/restaurants`	Employees (assignment), Payroll (Restaurant column), Performance (restaurant DPD), Delivery proximity gate
Zones	`zones`	`/zones`	Employees, Live Tracking filter, Payroll Zone Category, Order recon
Vehicle types	`vehicle_types`	`/settings/vehicle-types`	Vehicles form, Employees Vehicle column, Fuel limit default
Employee statuses	`employee_statuses`	`/settings/employee-statuses`	Employees form, HR Payroll Status paint, Payroll status pill
Employee sections	`employee_sections`	`/settings/employee-sections`	Employees form, approval gate (`is_operational`)
Nationality categories	`nationality_categories`	`/settings/nationality-categories`	Employees form, HR Payroll Nationality, eSign snapshot
Sponsorships	`sponsorships`	`/settings/sponsorships`	Employees form, HR Payroll Sponsor, eSign snapshot
Service types	`service_types`	`/settings/service-types`	Fleet Service records, maintenance programs
Payroll clients	`payroll_clients`	`/payroll/settings`	Combined Payroll, HR Payroll, Client name column

Cache invalidation is by **query-key factory** (`src/lib/query/query-keys.ts`) plus `revalidatePath` on the affected routes. There is no polling sweep: a save writes the catalog and invalidates the readers.

The sync bug this fixed. A new partner added in OperationsHub did not appear in the Employee Client dropdown because the dropdown read a module-local list. It now reads `partners` through the same server read the OperationsHub page uses. `source_companies` (Company) and `sponsorships` follow the same rule: every active row is listed, and a stored-but-inactive value is preserved rather than snapped to a default.

3.3 Payroll engines

Two engines, deliberately separate:

A. Payroll Hub month grid (`src/features/payroll/`)

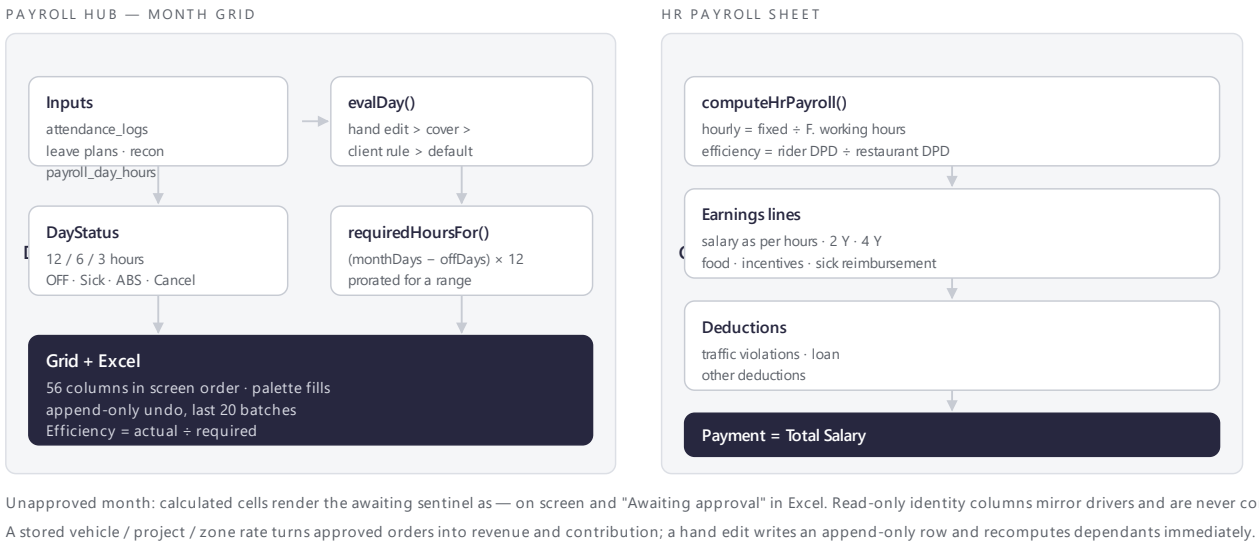


Figure 4. Two engines, deliberately separate: the month grid decides what a day was, HR Payroll prices the month. Neither recalculates the other.

Concern	Rule	File
Work day unit	`PAYROLL_DAY_HOURS = 12`	`payroll-formulas.ts`
Default OFF	`PAYROLL_DEFAULT_OFF_DAYS = 2` per month	same
Required hours	`(monthDays − offDays) × 12` , prorated for a partial range	`requiredHoursFor` , `requiredHoursForRange`
Day precedence	hand adjustment > Operations-approved cover > client rule > client default > legacy fallback	`payroll-rules-engine.ts`
Cell label	numeric (`12` , `6` , `3` , `4.3`) for hours; words for `OFF` / `Sick` / `ABS` / `Cancel`	`dayGridLabel`
Zone efficiency	previous **completed** month: `orders ÷ rider_days` vs fleet average; Good ≥ 110, Average ≥ 70, Low < 70	`payroll_zone_metrics`

Concern	Rule	File
Undo	append-only batches, `reason: 'undo'`, last 20	`payroll_manual_adjustments`
Export	56-column Excel matching the screen exactly, palette fills included	`combined-xlsx.ts`

B. HR Payroll sheet (`src/features/payroll/hr-payroll-formulas.ts`)

Client-specified field rules (review §3.4), implemented verbatim in `computeHrPayroll`:

Field	Rule
Payment	`= Total Salary`
Hourly Rate as per F. Working Days	`Fixed Salary ÷ F. Working Hours`
Efficiency	`Rider DPD ÷ Restaurant DPD` (0 when the denominator is 0)
Salary as per Hours	`Hourly Rate × Actual Working Hours`
2 Years Increment	`2-Year Applicable ÷ F. Working Days × Working Days`
4 Years Increment	`4-Year Applicable ÷ F. Working Days × Working Days`
Sick Leave Reimbursement	`Work Permit Salary ÷ 26 × Sick Leave Days`
Total Salary	`Salary as per Hours + 2Y + 4Y + Incentives + Food Allowance + Sick Reimbursement + Pending Salary + Total Allowances − Traffic Violations − Loan Deduction − Total Deductions`
Below-contract highlight	`hrSalaryShortfall()` returns a positive gap → row painted red when `Total Salary < Work Permit Salary`
Unapproved month	Cells that depend on Ops approval render `HR_AWAITING` (`\u0000awaiting`) → `—` on screen, "Awaiting approval" in Excel

Important separation: "From employees" columns (Nationality, Status, Work Start, Job End) are **read-only reflections** of `drivers`; only the Configuration and Calculated blocks are computed or editable. A hand edit is written through `adminSaveHrPayrollCells` and recomputes the dependent cells immediately.

C. Per-delivery incentive (`functions/src/core/incentive.ts`)

```
target  = restaurant rule → zone rule → partner rule → rider's own zone/partner
bands   = tier rates applied to consecutive bands above the target
gate    = compute_incentive_amount honours the DPD target for
          EVERY reward_mode / payout_mode / overrides_others
history = band math starts at app_settings.incentive_band_math_from
```

3.4 Inline editing, optimistic UI and read-through caching

Mechanism	Where	Behaviour
Optimistic mutation	`updateDeliveryStatus`, payroll cell edits, notification templates	Patch the cache first, then commit; a failed side-effect (earnings recalc) never discards the write
Append-only edit log	`payroll_manual_adjustments`	Every hand edit stores `before`, `after`, `reason`, `actor_id`; undo is a new row, never an update
Autosave	Payroll cell edit, HR cell edit	Debounced; a toast only on add/delete/reset
Read-through snapshot	`functions/src/router/read-snapshot.ts`	Scan-lane results are gzipped into `page_read_snapshots`, keyed by `fn
Epoch-keyed caches	`epoch-cache.ts`, `config-cache.ts`	Bounded LRU per isolate; invalidated by counter, not by TTL alone
Previous data held	`placeholderData: keepPreviousData` on every list	Changing a filter does not blank the table

3.5 Global search, filters and clickable KPI cards

Capability	Implementation
Row search	`fieldSearchMatch(row, query, accessors)` — one matcher across list pages; surname, code, restaurant, `EMP` digit run
Per-column filters	`ComparisonColumnFilters` / `PayrollColumnsMenu` / `TableToolbar` — text contains, value list with explicit **All**, numeric **range** (min/max), date range
Group by	`groupRows()` + `useGroupedListView()` — groups start closed, header shows value + count, header checkbox selects the whole group across its pages
Clickable KPI cards	`KpiCard` accepts `onClick` + `selected`; the tile applies the filter and the table, charts and sibling tiles re-derive from the same selector. Client portal uses `riderCountsByWidget` / `filterRidersByWidget` so a tile and the table it opens cannot disagree
Table toolbar	`TableToolbar` = Search + Group by + List/Grid + Columns, all URL-backed where sensible and persisted per user in `admin_ui_preferences`
Numeric-only guard	MG ID / Civil ID filters strip non-digits before the predicate (`sanitizeNumericFilter`)

3.6 Domain gates that carry business rules

Gate	Rule	File
Delivery proximity	In range when inside the assigned **zone polygon or any assigned store range**; store pin wins over polygon	`DeliveryProximityService.evaluate`, `isWithinDeliveryRange`
`zoneDescribesWorkArea`	A zone holding none of the rider's stores is **withheld** from the app, so the store range decides	`functions/src/rpcs/driver-deliveries.ts`

Gate	Rule	File
Zone idle timeout	45 min outside **zone or store range** → auto checkout; 20 min return grace after a delivery	`zone_monitor_provider.dart`, `admin_run_attendance_auto_checkout`
Pickup order ID	Digits only, 1–32; unique **per store per Kuwait day**, not fleet-wide	`driverCreatePickup`
Stale pickup	`app_settings.pickup_auto_cancel_hours` (default 6) → cron cancels and notifies	`/api/cron/expire-stale-pickups`
Driver level D	Month-to-date: `activeDays ≥ 3` and `verified < activeDays × target` (fallback 25)	`functions/src/rpcs/driver-levels.ts`
Level D override	Admin override writes a notification with deep link `musallam:///profile/performance-level`	same
Accident case	Frozen vehicle + driver snapshot so a later plate change cannot rewrite history	`vehicleAccidents`
Maintenance program	One row per **vehicle × program**; no approved rule ⇒ `no_rule`, not a guess from defaults	`service-maintenance-model.ts`
Odometer	Only ever moves **up** on unattended paths; a lower reading needs a written reason	`odometerDecision`, `recordManualOdometer`
E-sign stamp	Named slot per signer, never "last page bottom-right"; `sheet_render_version` gates measured boxes	`esign-signature-slots.ts`
Request SLA	`sla_minutes` per step; the hourly sweep **raises attention, never decides**	`admin_run_request_sla_sweep`

4 Cloud Functions, Serverless Endpoints & Background Jobs

4.1 Callable routing — three lanes, one cold start

All callables live in **one** Firebase functions package and are reached through three exported `onCall` routers plus the legacy names. `functions/src/router/lanes.ts` is the single source of truth, shared by the functions package and the browser bundle.

Lane	Export	Region	CPU / Mem	Concurrency	Instances	Timeout	Serves
light	`adminApi`	`me-central1`	1 vCPU / 2 GiB	8	0–8	60 s	Every admin callable not on scan/heavy
scan	`adminApiScan`	`me-central1`	1 vCPU / 4 GiB	2	0–6	**180 s**	19 unbounded readers (lists, payroll snapshots, attendance, fleet snapshot, order comparison…)
heavy	`adminApiHeavy`	`me-central1`	1 vCPU / 4 GiB	1	0–3	300 s	Purges (17), earnings recalculations, e-sign compose, import chunk claim, level sweep

`LANE_CLIENT_TIMEOUT_MS` keeps the browser abort ≥ the server timeout: `light 60 s`, `scan 180 s`, `heavy 300 s`. `laneOf(name)` decides from the registry, so a new callable cannot be shipped on the wrong lane; `assertPartition()` fails the build if a name is on two lanes or missing from the exports.

Scan-lane membership (19): `adminPerformanceOpsSnapshot`, `adminOrderComparisonSnapshot`, `adminOrderReconCompare`, `adminDeliveriesStatusCounts`, `adminDeliveriesCountsByFilters`, `adminListDriversPage`, `adminDriversFilterValues`, `adminLiveFleetSnapshot`, `adminPayrollRuleSnapshot`, `adminPayrollMonthSnapshot`, `adminHrPayrollSnapshot`, `adminHrAppHours`, `adminListAttendanceDaily`, `adminAttendanceKpis`, `adminIncentiveDailyReport`, `adminEmployeesDashboard`, `adminOperationsDashboard`, `adminClientPortalDashboard`, `adminListDriverLevels`.

4.2 Callable inventory — 319 exports across 73 modules

Module file	Exports	Representative names
`drivers.ts`	7	`adminListDriversPage`, `adminDriversFilterValues`, `adminUpsertDriver`, `adminArchiveDriver`
`drivers-admin.ts`	8	`adminApproveDriver`, `adminCreateDriverIntake`, `adminRestoreDriverIntake`
`driver-home.ts`	3	`driverGetHomeDashboard`, `driverSetDutyState`, `driverCheckInContext`
`driver-deliveries.ts`	5	`driverCreatePickup`, `driverCompleteDelivery`, `driverCancelDelivery`, `driverGetDeliveryProximityContext`
`driver-location.ts`	2	`driverReportLocation`, `driverClearLiveLocation`

Module file	Exports	Representative names
`driver-earnings.ts`	4	`driverGetEarningsSummary`, `driverGetExtraEarnings`, `driverGetEarningsDetail`
`driver-levels.ts`	4	`driverGetMyLevel`, `adminListDriverLevels`, `adminSetDriverLevelOverride`, `adminEvaluateDriverLevelsSweep`
`driver-requests.ts`	7	`driverCreateRequest`, `driverClarifyRequest`, `driverAcknowledgeRequest`, `driverRespondReschedule`
`driver-esign.ts`	8	`driverListEsignRequests`, `driverGetEsignRequest`, `driverSubmitEsignature`, `driverDeclineEsignature`, `driverMarkEsignViewed`, `driverAddEsignAttachment`
`driver-visits.ts`	3	`driverListVisitSlots`, `driverBookVisit`, `driverCancelVisit`
`driver-vehicle.ts`	2	`driverGetAssignedVehicle`, `driverReportFuelFill`
`driver-devices.ts`	6	`driverReportDeviceMeta`, `driverRegisterDevice`, `driverReleaseDeviceSession`
`driver-push-tokens.ts`	2	`driverUpsertPushToken`, `driverDeactivatePushToken`
`driver-telemetry.ts`	1	`driverIngestTelemetry` (≤100 events, `event_id` idempotency)
`driver-lists.ts`	10	`driverListMyDeliveries`, `driverListNotifications`, `driverListTenureOptions`
`driver-shift.ts` / `driver-attendance-get.ts` / `driver-pay.ts` / `driver-security.ts` / `driver-profile.ts` / `driver-auth.ts`	2 / 1 / 2 / 1 / 3 / 3	`driverSubmitDailyShift`, `driverGetTodayShift`, `driverGetAttendance`, `driverGetPayCalculator`, `driverLogSecurityEvent`, `driverUpdateAvatar`, login/passcode
`accidents.ts`	2	`driverReportAccident`, `driverListMyAccidents`
`payroll.ts`	1	`adminPayrollRuleSnapshot`
`payroll-more.ts`	7	`adminPayrollMonthSnapshot`, `adminSavePayrollIdentity`
`payroll-edits.ts`	6	`adminApplyPayrollAdjustments`, `adminSetPayrollColumnConfig`
`payroll-clients.ts`	9	`adminListPayrollClients`, `adminSavePayrollClient`, `adminDeletePayrollClient`
`hr-payroll.ts`	3	`adminHrPayrollSnapshot`, `adminSaveHrPayrollCells`
`attendance-ops.ts`	7	`adminListAttendanceDaily`, `adminAttendanceKpis`, `adminCorrectAttendance`, `adminRunAttendanceAutoCheckout`
`attendance.ts`	2	`adminListAttendanceRequests`
`requests-core.ts`	6	`adminListRequests`, `adminGetRequest`, `adminCreateRequest`
`requests-workflow.ts`	9	`adminDecideRequest`, `adminForwardRequest`, `adminRunRequestSlaSweep`, `adminAutoCloseRequests`

Module file	Exports	Representative names
`esign-requests.ts`	12	`adminListEsignRequests`, `adminGetEsignRequest`, `adminCreateEsignRequest`, `adminRemindEsignRequests`, `adminExpireEsignRequests`
`esign-batches.ts`	11	`adminCreateEsignBatch`, `adminProcessEsignBatchChunk`, `adminRetryEsignBatchRow`
`esign-signers.ts`	8	`adminListEsignSigners`, `adminAddEsignSigner`, `adminReorderEsignSigners`
`esign-compose.ts`	1	`esignComposeSignedDocument` (heavy)
`visits.ts` / `visits-shared.ts`	5	`adminListVisits`, `adminUpdateVisitStatus`, `adminRescheduleVisit`, `adminSetVisitNoteToRider`
`operations-dashboard.ts`	4	`adminOperationsDashboard`, `adminClientPortalDashboard`, `adminListLeavePlans`
`performance-list.ts`	2	`adminListDriverPerformance`, `adminDriverPerformanceDaily`
`performance-extra.ts`	6	`adminPerformanceOpsSnapshot`, `adminDpdEfficiencySnapshot`, `adminPerformanceTrend`, `adminPerformanceOpsBounds`
`performance-ratings.ts`	6	`adminListDriverPerformanceRatings`, `adminUpsertDriverPerformanceRating`, `adminSetPerformanceTeamMember`
`rules.ts`	11	`adminListDeliveryRules`, `adminUpsertDeliveryRule`, `adminListIncentiveRules`, `adminUpsertIncentiveRule`
`catalogs.ts`	21	`adminUpsertVehicleType`, `adminUpsertVehicleCompany`, `adminUpsertEmployeeCatalog`, `adminUpsertRestaurant`, `adminUpsertPartner`, `adminUpsertZone`, `adminUpsertServiceType`
`recon-reports.ts`	5	`adminOrderComparisonSnapshot`, `adminOrderReconCompare`, `reconcileDeliveryVerification`, `reportDeliveryOrders`
`notifications.ts`	2	`adminCreateCampaign`, `adminSendCampaign`
`notification-audience.ts`	4	`compileNotificationAudience`, `estimateNotificationAudience`, `resolveImportDriverIds`
`fleet.ts`	2	`adminLiveFleetSnapshot`, `adminGetDriverDayRoute`
`fleet-worker-http.ts`	3	`workerFleetRead`, `adminIngestDriverPositions`, `adminRecordFleetEvents` (onRequest + `X-Worker-Secret`)
`day-route.ts`	1	`adminGetDriverDayRoute`
`purge.ts` / `purge-all.ts` / `purge-entities.ts`	5 / 12 / —	24 entities, `_admin_purge_require(slug)` gate, FK release before delete
`retention.ts`	5	`cleanupDriverLocationEvents`, `cleanupDriverOperationEvents`, `cleanupDriverTelemetryEvents`, `cleanupStaleDriverLocations`

Module file	Exports	Representative names
`service-odometer-sync.ts` / `service-distance-sweep.ts`	1 / 1	scheduled, see §4.4
others	—	`fuel.ts`, `payouts.ts`, `earnings.ts`, `driver-list-plan.ts`, `misc-admin.ts`, `sync-staff-claims.ts`, `home-snapshot-repair.ts`, `leave-plans.ts`, `loan-tenure.ts`, `order-exceptions.ts`, `employee-dashboard`, `deliveries-counts/status`, `check-in-gate`, `delivery-context*`, `bites-*`

4.3 Next.js API routes — 49 `route.ts` files

Group	Routes	Method	Auth	Purpose
Cron (13)	see §4.5	GET/POST	`CRON_SECRET`	Scheduled sweeps
Order recon	`/api/order-recon/{preview,commit,comparison,undo,redo}`	POST	session + `order_recon.*`	300 s budget; Excel melt, compare, replay
Driver uploads	`/api/driver-uploads/{presign,confirm,proxy,read,mine}`	POST/GET	rider session	R2 presign → PUT → confirm; proxy streams when S3 signing is unavailable
eSign	`/api/esign/{document-download,batch-zip,render-bench}`	GET/POST	`esign.sign` or `requests.manage`	Same-origin streaming so an expired JWT cannot modal an error
Deliveries	`/api/deliveries/proof-download`, `/api/deliveries/[id]/gps`	GET	`deliveries.view`	Attachment download; day route
Driver app	`/api/driver-app/{active-release,firebase-config,notifications,notification-media}`	GET/POST	anon + rider token	Force-update gate, config, inbox
Admin ops	`/api/admin/{app-releases/*,driver-documents/*,locales/*,translations/*}`	GET/POST	`settings.manage`	Release upload, doc upload, locale editor
Assistant	`/api/assistant/chat`	POST	`assistant.view`	AI Gateway (`openai/gpt-4o`), 8 steps, read-only tools
Misc	`/api/build-id`, `/api/storage/signed-url`, `/api/launcher/weather`, `/api/live-tracking-v2/token`, `/api/menu-icons/[file]`, `/api/notifications/events`, `/api/drivers/import-template.xlsx`	GET/POST	mixed	Version gate, signed URLs, weather (Open-Meteo, 30 min), V2 socket ticket, uploaded menu icons, engagement beacon, template download

4.4 Cloud Functions — scheduled (`onSchedule`, 5)

Function	Schedule (Kuwait)	Purpose
`serviceOdometerSync`	`0 0,6,12,18 * * *`	Dashcam odometer pull; reads `service_odometer_sync/current`; secret `SERVICE_ODOMETER_APP_SECRET`; never rewinds a reading
`serviceVehicleDistanceSweep`	`45 23 * * *`	Nightly bike day-distance for riders who never clocked out
`deliveryProximitySweep`	periodic	Rebuilds the shared delivery-context document (fleet settings, zones, restaurants + geofences)
`homeSnapshotRepair`	periodic	Repairs `driver_home_daily` rows
`bitesMirrorSweep`	periodic	Bites order mirror reconciliation

4.5 Vercel cron jobs (`vercel.json`, 13)

Path	Cron	Purpose
`/api/cron/notifications-dispatch`	`*/5 * * * *`	FCM dispatch for scheduled campaigns
`/api/cron/notifications-automations`	`*/5 * * * *`	Lead-day automations (document expiry, visits)
`/api/cron/attendance-auto-checkout`	`*/5 * * * *`	45-min zone exit, shift end, orphaned duty
`/api/cron/dpd-shift-notices`	`*/5 * * * *`	30-min warning, target reached, shift summary
`/api/cron/esign-batch-drain`	`*/5 * * * *`	Finishes a bulk e-sign batch after the tab closes
`/api/cron/locations-cleanup`	`*/2 * * * *`	Retire stale `driver_locations`
`/api/cron/expire-stale-pickups`	`*/15 * * * *`	Cancel a pickup stuck `in_transit`
`/api/cron/roster-alerts`	`*/15 * * * *`	No-login, late, early-drop, overtime
`/api/cron/requests-sweep`	`0 * * * *`	SLA breach attention + auto-close
`/api/cron/storage-uploads-cleanup`	`0 4 * * *`	Orphaned uploads
`/api/cron/driver-ops-retention`	`20 1 * * *`	30-day operation events
`/api/cron/driver-telemetry-retention`	`40 1 * * *`	14-day telemetry
`/api/cron/performance-rollup`	`0 2 * * *`	Rewrite trailing `driver_performance_daily` window

4.6 Cloudflare Worker `dpd-live`

Piece	File	Role
Worker entry	`infra/workers/dpd-live/src/index.ts`	Routes `/ingest`, `/ingest/ws`, `/ws`, `/latest`, `/fleet-stats`, `/history`, `/history/days`, `/archive/*`
`FleetRoom` Durable Object	`fleet-room.ts`	One room = whole fleet. Holds roster, zones, shifts, thresholds and a 10-minute trail buffer in memory; batches a durable flush every 10 s through `adminIngestDriverPositions`
Interest management	`fleet-interest.ts`	Viewport + filter culling before serialise — 500 riders affordable
Rules	`fleet-rules.ts`	Shared with `functions/src/rpcs/fleet-ingest-rules.ts`; hysteresis + grouped cooldowns for Class B events
GPS archive	`gps-archive.ts`	SQLite + R2 `dpd-gps-prod`; minute files compacted hourly per rider
Auth	`auth.ts`	Firebase JWKS verification, tri-state (`ok` / `rejected` / `unavailable`), 5-min negative cache only on a **definitive** rejection

Read-cost posture. `ROSTER_TTL_MS` 10 min; a rider not on the roster is read **individually** (`workerFleetRead { op: "driver_snapshot" }`, ~12 reads) with a 5-minute negative cache, instead of a whole-fleet snapshot per batch. `FLEET_BASE_CACHE_MS` 120 s. `SNAPSHOT_POLL_MS` 60 s on the client.

4.7 Webhooks and outward integrations

Integration	Direction	Transport	Notes
FCM	outbound	Admin SDK `sendEach`	Batched 500/topic; token per driver; stale token picks `max(last_seen_at)`
BDashcam / VSS GPS MEA	outbound (blocked)	HTTPS	`/sys/login` + `/position/realtime/vehicleGps`; **gateway rejects the login from our network — awaiting vendor allowlist**, config `enabled:false`
Open-Meteo	outbound	HTTPS	`/api/launcher/weather`, 30 min cache
AI Gateway	outbound	HTTPS	`/api/assistant/chat` only
Cloudflare `dpd-live`	inbound	`X-Worker-Secret`	`workerFleetRead`, `adminIngestDriverPositions`, `adminRecordFleetEvents`
Bites	both	`BITES_BRIDGE_SECRET`	`bitesFleetSocketTicket`, `bitesMirrorSweep`

4.8 Audit and observability

Concern	Implementation
Staff audit	`logAdminMutation()` on every write → `admin_activity_logs` with `before` / `after` and `page_path`; driver-document **view and download** are logged too

Concern	Implementation
Rider RPC audit	`log_driver_operation` in-transaction; `log_driver_operation_autonomous` over `dblink` for RPCs that raise (so a refused pickup is still recorded)
Callable logs	`writeRouterLog(lane, fn, inflight, ms, snapshot)` → single-line JSON per call
Error tracking	Sentry client + functions; `SentryUserSync` binds the staff id
Cost telemetry	`admin_cost_samples`, `scripts/admin-cost-audit.mjs`

5 Complete Feature Matrix

Module	User-facing features	Inline editing	Real-time / sync	Reporting & export
Dashboard / Launcher	Dark launcher grid, live Kuwait weather, date + clock, badges, ⌘K search	—	Weather 30 min; badges on poll	—
Control Tower	Cross-module KPI snapshot	—	60 s poll	—
Live Tracking V1	Map pins by status, driver popup with address, recent orders, filters, fullscreen	—	15–30 s poll + `driver_locations` realtime	Day route, CSV
Live Tracking V2	1 Hz deck.gl vehicle sprites, 10-min trails, playback scrubber, floating rail + insights, collapsible overlays	—	WebSocket 4 Hz → 1 Hz poll fallback; interpolated on client	Excel
Employees	List / card view, 40+ columns, tabbed detail (Private info, Documents, Attendance, Assets, Earnings, Loan, Performance, Diagnostics), approve / verify, archive, delete, bulk import + export, custom columns	Cell edit, quick-edit, per-column filters, `custom_fields`	Catalog-driven dropdowns; badge poll	XLSX import (parity-tested) + filtered export
Employee Groups	Create, member add/remove, Excel import with mismatch rejection	Member chips	—	CSV template
Fleet — Vehicles	KPI strip (Unassigned/Bike/Car/Active/Replacement), 18 columns, identity header, Assign rider, Assets tab, 5 ledgers	Per-column filters, bulk assign / status / company	Assignment reflected on Employees + Payroll	Sample XLSX + export + import batches with undo/redo
Fleet — Fuel	Daily/Weekly/Monthly/Custom (≤31 Kuwait days), >90% red, fuel + refund requests, per-driver page	Withdrawn override per rider/vehicle/month	Request queue polls	Weekly/monthly export
Fleet — Service	Records (5-stage workflow), Cost analysis (KPI, monthly Contract/Service chart, 4 breakdowns), Maintenance (vehicle × program), Bikes page	Odometer dialog, stage change, baseline import	Program rules resolve live	Styled Excel per tab, baseline + bike import
Fleet — Accident	Case list + 5-stage, frozen vehicle/driver snapshot, costed repair, attachments, detail page, rider-filed cases	Stage, cost, documents	Rider report → inbox	Excel
Fleet — Handover / RJ	Handover ledger, RJ (rejected-request) list	Record dialog	—	Export

Module	User-facing features	Inline editing	Real-time / sync	Reporting & export
Deliveries	Filters (status, zone, partner, date, flags), bulk verify/reject, detail sheet with proof, GPS trail, Orders Report, reconciliation compare	Status change (optimistic)	Realtime on new rows; 60 s counts	Orders Report matrix XLSX, recon 6-sheet report
Order Exceptions	Rider-filed rejected/cancelled list, reason + proof count, review dialog with signed URLs	Review approve/reject	Rider push on decision	Summary + XLSX
Payroll Hub — Summary	5 KPI cards, status row with counts, 15-row grid, column picker	Identity cells, status pill (custom label)	Snapshot 5 min; epoch-busted on write	Per-tab Excel, CSV
Payroll — Combined	Spreadsheet grid: 7 identity cols, 31 day cols, ordered summary; drag select, Ctrl+C/V/D, fill handle, paste tiling	Every cell; reason required for hand edit; undo/redo	Optimistic + append-only log	56-column Excel identical to screen
Payroll — Attendance & Orders	Dual sub-columns (Payroll / Orders) per day, order override input	Day cell + order count	Recon/recon-override wins	CSV
Payroll — Application Attendance	Per-day app usage, active days, average hours	—	—	CSV
HR Payroll	50 columns in client order, Summary (4 cards + company table), Payroll, Configuration	Configuration + HR cells; calculated block read-only	Awaiting-approval state per month	Excel + import of allowances/deductions
Payroll Settings	Client criteria, hour values, rule builder, "try the rules", zone efficiency table with overrides, change log	Rules, thresholds, zone overrides	Autosave debounce	—
Attendance	Live presence, logs, exceptions, analytics, driver explore sheet, correction	Correct check-in/out and status	60 s; realtime on `attendance_logs`	CSV
Shift Attendance / Roster	Setup, weekly Excel upload, schedule changes, alerts (ack with outcome), report	Ack outcome	15-min alert cron	Excel with scheduled/worked hours
Operations Hub	Hub tiles + two-panel Figma shell	—	—	—
Notifications	Campaign list with audience counts, 4-step create wizard, templates, automations, history, analytics, per-recipient drill-down	Template body, action fields	Dispatch every 5 min	Engagement report, export

Module	User-facing features	Inline editing	Real-time / sync	Reporting & export
Roles & Permissions	3-pane access editor (280 / 1fr / 300), Staff rail with dept chips + counts, 4 access levels, sub-view chips, launcher preview, Manager↔User with full-catalog seed, role select, copy access	Ticks, role, level	Save bumps `auth_epochs.access`	—
Menu Editor	Per-role sidebar tree, drag, hide, rename, reset	Tree edits	Saved config is the sidebar	—
Languages	Locale table + translation editor writing `src/messages/*.json`	Per-key editor	—	Parity guard via `test:i18n`
Employee Desk — Outgoing	Template tiles (8 categories + penalty sub-menu), bulk send, tracked sends with progress, admin tools	—	Tracker refreshes on resume	Batch ZIP, unsigned PDF
Employee Desk — Incoming	8 queues with counts, needs-action list, filters, sort, upload, forward/escalate, confidential mask	Decision, forward, comment	Push on decision	Per-type queue export
eSign	Template library + builder, send / bulk (Excel), To Sign inbox, batch detail, remind (cooldown), decline, signed copy, signature proof	Field map, signer order, sheet HTML	Drain cron finishes a closed tab	A4 PDF, signed PDF, signature PNG
Visit Slots	Hub by department, All visits with header filters, Calendar Day/Week/Month/Year, Reception check-in, Slots, Departments & desks, Branches, Reports	Status, reschedule, note to rider	KPI windows server-side	Excel + QR ticket
Performance	5 SOP tabs, 8 slicers, Client theme (Americana blue / Keeta gold), Order Status slicer, sortable + filterable rider table, 60 s refresh	—	60 s snapshot	Per-chart CSV, rider XLSX
Performance Settings	Component weights with 3-way preview (old / saved / new), team membership	Weights, members	Save re-scores the fleet	—
Client Portal V2	Partner-scoped dashboard, 6 clickable KPI tiles → rider table with pager, leave chart, joiners, employee list with documents	—	2-min snapshot	CSV, document download
Partner Preview	Super admin "view as Keeta / Americana" with exit banner	—	—	—
Settings — Branding / App	Name, logo, font, maintenance, driver-app maintenance, force-update minimum version	Forms	Epoch-busted	—
Settings — Catalogs (13)	Vehicle types/uses/companies, employee statuses/sections, nationality, bank codes, job positions, sponsorships, source companies, service types, payroll clients	Row edit; some bulk import	Every form reads the catalog live	Sample XLSX where applicable
Settings — Custom columns	**Super Admin only.** Add columns per entity (Employees, Clients, Restaurants, Zones, Vehicles); a `select` / `checkbox` / `date` field also becomes a slicer	Definition editor	Definitions invalidate every list read	—

Module	User-facing features	Inline editing	Real-time / sync	Reporting & export
Settings — Data cleanup	Clear all per module (typed confirmation), Clear by filter, blockers shown first	—	—	Storage key manifest
Driver Devices	Build / phone / profile / last seen / severity, Force update, notify	Bulk force update	Last-seen poll	XLSX
Document Expiry	Expired / 1 week / 1 month / 3 months, per-doc lead days	Expiry date, lead days	Automations notify	CSV
Compliance	Invalid / rejected / cancelled day upload, rider-day ledger	Upload	—	—
Wrong Actions	Incident list, detail, KPIs by severity, weight feeds Performance	Create / edit	—	CSV
Earnings / Payouts	Daily earnings, wallet ledger, payout runs + detail	Adjust, recalc range	—	CSV
Support	Threads, messages, SOS, appointments	Reply	Poll	—
Staff Assistant	Chat over existing reports (EN + AR, RTL), 8-step tool budget, read-only	—	—	Excel from the same numbers
System	Cloudflare R2, storage, activity logs, access requests, data cleanup, menu editor, languages	—	—	Log export

5.1 Cross-cutting capabilities present on every list module

Capability	Mechanism
Column order, visibility, sort, group, view	`TableToolbar` + `useTablePrefs` → `admin_ui_preferences` (per user) with role defaults
Search	`fieldSearchMatch` ; ≥4-char prefix on indexed fields, full scan fallback while the backfill is incomplete
Numeric range filter	Min/max on every numeric column
Group by	`groupRows` + `useGroupedListView` ; groups closed by default, header selects the group across its pages
Pager	`from–to / total` , 80 rows a page, per-group pager inside an opened group
Row → detail	Row click opens `/resource/[id]` ; edit is a modal only (`?add=1` , `?edit=1`)
Clear all	`ClearAllModuleButton` gated on `*.bulk_delete` , typed confirmation, FK release order
Export	Excel/CSV reflecting the exact visible columns and order
Back button	Every detail route carries a Back link resolving through a per-module allowlist

6 Appendices

6.1 Environment variables (names only)

Group	Variables
Firebase Admin	`FIREBASE_PROJECT_ID`, `FIREBASE_CLIENT_EMAIL`, `FIREBASE_PRIVATE_KEY`, `FIREBASE_SERVICE_ACCOUNT_JSON`, `FIREBASE_API_KEY`
R2	`R2_ACCOUNT_ID`, `R2_ACCESS_KEY_ID`, `R2_SECRET_ACCESS_KEY`, `R2_BUCKET_NAME`, `R2_ENDPOINT`
eSign worker bridge	`ESIGN_R2_WORKER_URL`, `ESIGN_R2_WORKER_SECRET`
Worker	`WORKER_SHARED_SECRET`, `FLEET_WS_URL`, `FLEET_WS_TOKEN_SECRET`
Client-exposed	`NEXT_PUBLIC_GOOGLE_MAPS_API_KEY`, `NEXT_PUBLIC_GOOGLE_MAPS_MAP_ID`, `NEXT_PUBLIC_FIREBASE_*`
Cron	`CRON_SECRET`
Feature switches	`ADMIN_API_ROUTER`, `PAGE_READ_SNAPSHOT`, `GPS_SINK`, `GPS_DUAL_WRITE`, `LOCATION_EVENTS_ARCHIVED`
Secrets (Secret Manager)	`SERVICE_ODOMETER_APP_SECRET`

6.2 Test suites (27 `npm run test:*` scripts)

`test:geo` · `test:zones` · `test:drivers` · `test:permissions` · `test:i18n` (**12,625 keys × 2**) · `test:service` · `test:storage` · `test:assistant` · `test:recon` (57) · `test:vehicles` · `test:driver-devices` · `test:auth` · `test:performance` · `test:payroll` · `test:dpd` · `test:deliveries` · `test:earnings` · `test:visits` · `test:esign` · `test:templates` · `test:employeedesk` · `test:requests` · `test:compliance` · `test:notifications` · `test:attendance` · `test:fleet` (308) · `test:accidents` · `test:portal` · functions `npm test` (496).

6.3 Known constraints and open items

Item	State
Supabase Postgres	**Paused.** 398 migrations retained as history; nothing reads them
BDashcam odometer feed	Endpoint configured, secret stored, function deployed — **paused** because the vendor gateway rejects `/sys/login` from our network (see `scripts/probe-bdashcam-api.mjs`)
Fleet Service maintenance	Rules engine shipped; **0 maintenance rules seeded in production** until `scripts/service-maintenance-seed.mjs` is applied and the baseline import re-run
Custom columns	Live for **Employees** and **Clients**; Restaurants, Zones and Vehicles are registered but not yet wired (`hasListColumns: false`), so the settings screen does not offer them
Numeric-only Excel for day cells	`combined-xlsx.ts` writes real numbers for hour cells; the CSV paths still emit strings

Item	State
`driver_app_force_update`	Server switch live; takes effect only once a build ≥ the configured `versionCode` is on Play
Local `next build`	Fails in this checkout while the parallel `operations-dashboard-v2` module is untracked and incomplete; Vercel builds from the committed tree

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End of document.